

(h)

Row argmax

Attention A

Grid BCCB

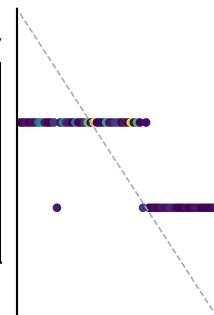
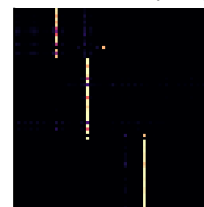
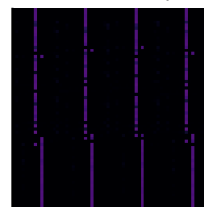
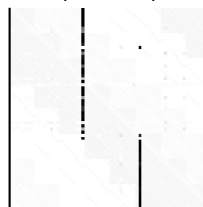
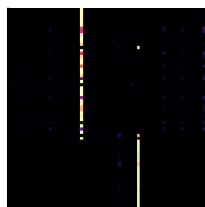
 $|A - \text{Grid}|$

Best flat BCM+perm Monarch-like proxy

ViT L0 H0

Grid err=0.98, out=1.10

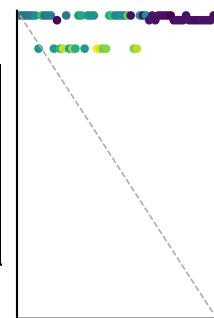
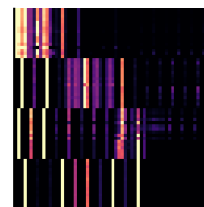
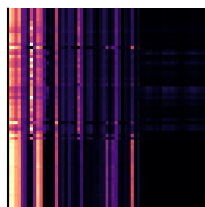
Perm err=0.86, Proxy err=0.67



ViT L5 H0

Grid err=0.84, out=0.33

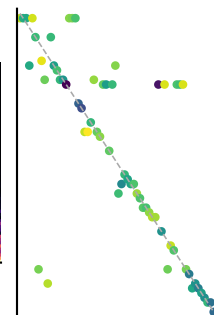
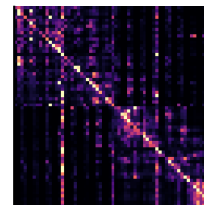
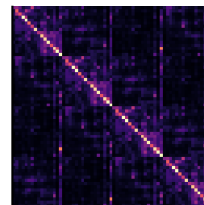
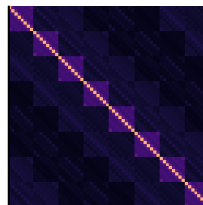
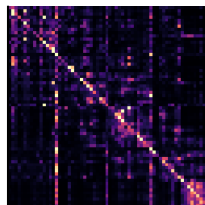
Perm err=0.43, Proxy err=1.23



Qwen L8 H0 F1

Grid err=0.70, out=0.47

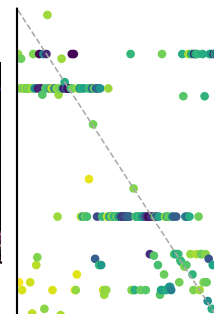
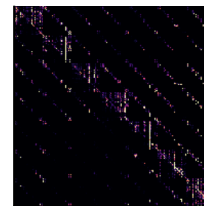
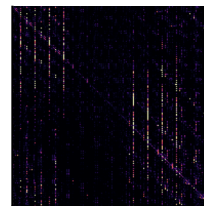
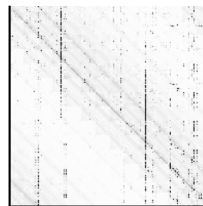
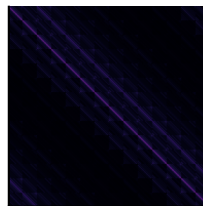
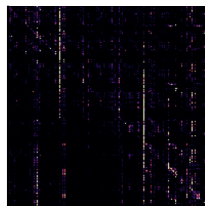
Perm err=0.60, Proxy err=0.24



Qwen L26 H0 F0

Grid err=0.98, out=0.72

Perm err=0.85, Proxy err=0.83



If grid-BCCB held, rows would look like cyclic shifts and row-argmax points would form offset diagonals. Actual maps show sinks, sparse rows, and content-dependent row patterns.